

ALPACA AI TRADING AGENTS HACKATHON

Devil's Advocate

One model proposes. A second attacks. One revision is allowed. Deterministic code decides what authority survives.

Frozen policy: **proceed** < 0.60 · **revise** 0.60–0.70 · **reject** > 0.70 · max 3 contracts/trade · fresh-state execution gate before broker submission.

What the system proved

STORED DECISION SESSIONS

9

7 included model revision

CODE SUBSTITUTIONS

3

safer alternatives generated by deterministic logic

OVERSIZED PROPOSALS CAUGHT

5

model intent constrained by hard authority limits

GATE TESTS

28/28

latest deterministic safety suite

Core result. The models are allowed to be imperfect. They may disagree, misclassify a risk, or propose too much size. They still cannot exceed the authority encoded in the gate. No-trade remains valid, and repeated disagreement never forces execution.

Calibration evidence

BENCHMARK CASES

30

FLAWED INTERVENTION

95.8%

8/28/26, 3:40 PM

Devil's Advocate — Judge Brief

24 flawed + 6 clean

23/24 revised or blocked

CLEAN IMMEDIATE PASS

83.3%

5/6 proceed immediately

SEVERITY AUC

0.951

flawed vs clean severity separation

Calibration result. Severity was a much stronger signal than exact failure-mode naming. With the frozen thresholds, 23/24 flawed cases triggered intervention while 5/6 clean controls proceeded immediately. Exact mode match was 12/24 (50.0%), so the system treats the label as diagnostic context rather than the safety decision itself.

INJECTED MODE	EXACT LABEL	MEAN SEVERITY	OBSERVED ADVERSARY LABELS
assignment risk	3/3	0.717	assignment_risk: 3
concentration	0/3	0.700	position_conflict: 3
insufficient premium	3/3	0.633	insufficient_premium: 3
overconfidence	0/3	0.567	insufficient_premium: 3
poor liquidity	3/3	0.717	poor_liquidity: 3
position conflict	0/3	0.650	insufficient_premium: 1, concentration: 2
recency bias	1/3	0.600	insufficient_premium: 2, recency_bias: 1
unfavourable expiry	2/3	0.717	unfavourable_expiry: 2, insufficient_premium: 1

Calibration source: data/calibration/adversary_20260828_182652.json

Three decision stories

These are the highest-value episodes from the audit trail: one safety substitution, one genuine live no-trade decision, and one revision that materially changed the trade.

data/decisions/2026-08-28-01-live-dry.json

2026-08-28 · session 1 · LIVE-DRY

SUBSTITUTE

PROPOSED

4x covered call · XLE260918C00065000

REVISION

4x covered call · XLE260918C00063500

FINAL RULING

substitute → 2x XLE260918C00063500

BROKER

No order placed

CHALLENGE

insufficient_premium · severity 0.55

CHALLENGE AGAIN

assignment_risk · severity 0.50

FRESH EXECUTION GATE

passed

Initial proposal exceeded contract cap

Contract changed on revision

Final size reduced

► Rationale

data/decisions/2026-08-28-02-live.json

2026-08-28 · session 2 · LIVE

REJECT

PROPOSED

4x covered call · XLE260904C00063500

REVISION

2x cash secured put · XLE260904P00061000

FINAL RULING

reject

CHALLENGE

assignment_risk · severity 0.55

CHALLENGE AGAIN

insufficient_premium · severity 0.50

BROKER

No order placed

Initial proposal exceeded contract cap

Strategy changed on revision

Contract changed on revision

Resolved to NO TRADE

► Rationale

data/decisions/2026-08-28-05-live-dry.json

2026-08-28 · session 5 · LIVE-DRY

SUBSTITUTE

PROPOSED

CHALLENGE

4x covered call · XLE260904C00063000

insufficient_premium · severity 0.50

REVISION
4x covered call · XLE260904C00063500

CHALLENGE AGAIN
insufficient_premium · severity 0.50

FINAL RULING
substitute → 2x XLE260904C00063500

FRESH EXECUTION GATE
passed

BROKER
No order placed

Initial proposal exceeded contract cap

Contract changed on revision

Final size reduced

► Rationale

Real live decisions

LIVE SESSIONS

2

competition paper account; mode=live

LIVE NO-TRADE DECISIONS

2

inaction remained a valid output

LIVE BROKER SUBMISSIONS

0

actual paper orders accepted by broker

LIVE-DRY VALIDATIONS

6

real market data, intentionally no submission

LIVE and LIVE-DRY are intentionally separated. LIVE means the full paper-trading path was enabled. LIVE-DRY used real broker state and market data but intentionally stopped before order submission.

Compact audit ledger

RUN	MODE	INITIAL PROPOSAL	PRIMARY OBJECTION	FINAL OUTCOME
2026-08-28 session 6	LIVE-DRY	4× XLE covered call	assignment_risk · 0.55	REJECT reject
2026-08-28 session 5	LIVE-DRY	4× XLE covered call	insufficient_premium · 0.50	SUBSTITUTE substitute → 2× XLE260904C00063500
2026-08-28 session 4	LIVE-DRY	2× XLE covered call	assignment_risk · 0.60	REJECT reject
2026-08-28 session 3	LIVE-DRY	4× XLE covered call	insufficient_premium · 0.35	SUBSTITUTE substitute → 2× XLE260904C00064000
2026-08-28 session 2	LIVE	4× XLE covered call	assignment_risk · 0.55	REJECT reject
2026-08-28 session 2	LIVE-DRY	3× XLP covered call	assignment_risk · 0.65	REJECT reject
2026-08-28 session 1	LIVE	2× XLE covered call	assignment_risk · 0.55	REJECT reject
2026-08-28 session 1	LIVE-DRY	4× XLE covered call	insufficient_premium · 0.55	SUBSTITUTE substitute → 2× XLE260918C00063500

What worked — and what did not

Strongest behavior

Risk escalation

Flawed proposals were separated from clean controls well enough to calibrate intervention thresholds from observed severity rather than guesses.

Known limitation

Exact labels

The adversary matched the injected failure-mode label only 12/24. Concrete market risks were stronger than abstract reasoning-bias labels.

Interpretation. The adversary is useful as a calibrated escalation layer, not as an infallible classifier. Exact naming is secondary; deterministic code remains responsible for enforceable safety.

Authority boundary

Models may

Propose trades · criticize reasoning · suggest a revision · abstain.

Models may not

Bypass contract caps · create naked calls · exceed collateral or coverage · override the daily loss halt · remove the kill switch · bypass the fresh execution gate.

Generated 2026-08-28 19:40 UTC from local JSON decision/calibration artifacts. This report performs no model calls, broker calls, or network access.